Final Report

Data Mining for Software Engineering

(50 points)

Due: Check D2L for due date (Submit D2L)

Objectives:

- To document project components in the final report

Instructions:

- This is an individual submission
- The submission should be between 1200 and 1500 words.

For this submission, you will compile all your work from weeks 13 to 15 into your final project report. The tasks that you need to perform to complete this assignment are as follows:

1. Your report should contain the following sections:
    a. Introduction (i.e., description about your project, problem statement, motivation, and proposed solution).
    b. Background (i.e., literature search).
    c. Experiment Design (i.e., information about RQs, data sets, algorithms, etc., that you worked on in week 13)
    d. Experiment Procedure (refer to your work in week-14)
    e. Results and Discussions (Refer to a sample description below, i.e., within this document, to know how to document this section).
    f. Conclusions
    g. References
2. Submit this assignment in D2L before the due date.

Grading Rubrics

Task-1 (Preprocessing) 30%
Task-2 (Evaluation plan) 40 %
Task-3 (Experiment procedure) 30%

WRITING RESULTS AND DISCUSSIONS

For this part of your project, you will elaborate on various components of your projects (as listed below). The tasks that you need to document results and discussions section is as following:

1. For the research questions (RQ's) that you identified for your project, you need to describe your results. For your understanding about this task, a hypothetical example is considered that attempts to *automate the loan eligibility of a person* depending upon various constraints passed to the prediction models. These constraints are age, income, student status, other monthly expenses, zip code, and employment status, etc. Assume that for this example, the following were identified as part of your experiment design.
    a. **Research Question (RQ-1)**: What is the accuracy of our model at predicting the eligibility of a person for a loan?
    b. **Data Set**: Assume that the data set (e.g., XYZ) used for this problem consists of real world data and there are 30,000 data instances. The data also included roughly equal number of cases that were either 'approved' or 'denied'. The models is trained using 20,000 instances and the remaining 10,000 are used for testing the prediction.
    c. **Algorithm**: Assume your model used *Probability based algorithms* to predict the outcome.
    d. **Evaluation metrics**: Your model is going to be evaluated using accuracy, recall, and precision and compared against the results from using the regression models by various researches.

    For the above experiment design (*note that this experiment design is hypothetical and contains very little description about various elements*). You need to write the result section as following:

    # Results

    Description: The results are obtained using the probabilistic algorithm using the XYZ dataset. The results are evaluated using the RQ's identified earlier and described in XXX section (*you can refer to your previous assignment or the section in this assignment if re-writing those again*). The results are presented in Table 2, where the confusion matrix (refer Table 1) is used to identify the true-positives, true-negatives, false-positives, and false-negatives. This confusion matrix is used to calculate the results in terms of accuracy, precision, and recall. (**You can add more similar details depending upon experiment design of your project).** The results obtained are structured around the RQ's identified earlier and are as following:

Table 1. Confusion Matrix

	Actual (Yes)	Actual (No)
Predicted (Yes)	True Positive	False Positive
Predicted (No)	False Negative	True Negative

RQ-1: What is the accuracy of our model at predicting the eligibility of a person for a loan?

Using the confusion matrix (shown in Table 2), following are a few observations regarding the results. (**NOTE: Basically in this section, you will be observing the results data**)

a. The number of truly predicted instances are 8,500 out of 10,000 making the accuracy of our model to 85%.
b. There were a total of 1500 data instances that were incorrectly identified as false-negatives and false-positives.
c. It is also observed from the data that there are 5100 instances that are 'eligible' and 900 'ineligible' instances.

Table 2. Results

	Actual (Eligible = Yes)	Actual (Eligible = No)
Predicted (Eligible = Yes)	4,500	900
Predicted (Eligible = No)	600	4,000

2. Next, for the results section that you completed in step-2, you will write the discussion section. **This section basically provides the insights/reasoning about the results** obtained (i.e., the observations that you identified about results). The following write-up attempts to present how a discussion section can be written.

Discussion of Results

The discussion about the results and the major observations are presented in this section around the RQ's that are identified earlier for this project. The prominent insights and implications are described as following.
RQ-1: This RQ attempts to evaluate the performance of our model used for this project problem. The analysis is presented around the observations identified in results section earlier.

Observation-1 (Model accuracy of 85%) – The model predicted 85% of the instances accurately, which is a fair improvement over other researches presented in the literature which could only predict 79% correct instances (**NOTE: I just made this up**). One reason behind this improvement has been because of the use of probabilistic models to train our model over regression models used by the researches in literature. Probabilistic models are most suited for the type of data that is collected to study the underlying problem identified for this project. (**Next, you can also provide a reason as to why your model incorrectly predict the other 15% of the data instances**). Similarly, **you can present the reasons behind the observations from the results section and what are the implications** of these observations (*e.g., our model can be used as a substitution for regression based model, especially, to predict the loan eligibility*).